AnswerBot — AI-Powered FAQ and Customer Support Automation

Problem Identification

Customer support operations may involve large backlogs of repetitive inquiries, making it difficult to organize FAQ entries, classify topics, prepare precise answers, maintain up-to-date documentation, and identify relevant information quickly. The documented root causes include time-consuming manual answer drafting, limited semantic search visibility, inefficient documentation workflows, and lack of unified permission layers for private documentation.

Proposed Idea

Build a secure RESTful backend that centralizes users and FAQ content and adds Google Gemini AI for automated FAQ generation and intelligent question-answering support. The selected concept is AnswerBot, using Node.js, Express.js, MongoDB with Mongoose ODM, JWT authentication, bcrypt, and Google Gemini.

Key Ideas

- Secure registration and login using bcrypt password hashing and JWT authentication.
- Complete FAQ CRUD operations with ownership and authorization controls.
- Search FAQs using keywords and semantic matching for faster information retrieval.
- AI-powered FAQ generation that creates questions, answers, and topic/category suggestions.
- Category management for organizing FAQ content.
- Centralized error handling and Mongoose schema validation for reliable API responses.
- Modular MVC architecture separating models, controllers, routing, services, and database operations.

Target Users & Value

The primary users are customer support operations managers, content creators, authenticated users, and public consumers. AnswerBot centralizes FAQ management, reduces repetitive manual work, improves information discoverability, and provides AI-assisted content creation. The REST API can also support future web or mobile clients.

Selected Scope

Authentication and authorization, FAQ management and search, category management, AI-powered FAQ generation, schema validation, centralized error handling, MongoDB persistence, and JSON REST API responses. The detailed project document also defines separate roles for administrators, content creators, authenticated users, and public users.

Core Technology Stack

Layer	Technology
Runtime / Backend	Node.js, Express.js
Database	MongoDB, Mongoose ODM
Authentication	JWT, bcrypt
AI Integration	Google Gemini
API Testing	Postman / ThunderClient
Architecture	MVC / Modular RESTful API

Future Enhancements

The detailed project document focuses on the current FAQ automation and REST API scope. Future client applications and additional customer-support automation can be built on top of the API.